

Chanyeok Choi

Robot Engineer · MS in Applied AI, Hanyang University

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Profile

Robot engineer working on **critical-phase detection** for vision-language-action policies (preprint, under review). Broader interests include safe and generalizable robot learning, multi-agent reinforcement learning, and sim-to-real transfer. Previously built field-deployed service robots (auto-climbing mopping platform, autonomous delivery platform) — most of my intuition about real-world control and edge cases comes from there.

Education

Hanyang University , Applied Artificial Intelligence - Advisor: Prof. Youngmoon Lee - Research on safety-RL, poisoning attacks on multi-agent RL, and vision-language-action policy failure detection.	Seoul, South Korea Feb 2022 – present
Hanyang University , Robotics Advisors: Prof. Youngmoon Lee, Prof. Taejoon Park, Prof. Changsoo Han	Seoul, South Korea Mar 2018 – Feb 2022

Experience

Hanyang University , Graduate Research Assistant Robot learning research under Prof. Youngmoon Lee. Critical-Phase Detection (CPD): zero-shot failure detection for frozen π_0 / $\pi_{0.5}$ VLA policies on LIBERO manipulation. Leave-one-out F1 0.86–0.89 with no per-task reward (preprint, under review). Poisoning attacks on multi-agent RL: adversarial reward-poisoning study on multi-agent control, including taxi ride-sharing systems. Safety-RL: bridging the sim-to-real gap by allowing safe optimization and exploration during robot learning. Vision (HumanPose): hybrid PoseNet/SegNet approach for crowded scenes with overlap and occlusion.	Seoul, South Korea Mar 2022 – present 4 years 3 months
KITECH — Process Platform Research Division , Research Intern Researched AI methods for detecting defective products on high-speed conveyor belts.	South Korea Mar 2021 – Aug 2021 6 months
Hanyang University — CAI Lab , Research Intern IKEA Furniture Assembly Collaborative Robot: RL simulation in which two manipulators identify randomly placed furniture parts and assemble them collaboratively.	Seoul, South Korea July 2019 – Feb 2020 8 months

Publications

Critical-Phase Detection for Vision-Language-Action Policies

Zero-shot per-step failure detector for frozen π_0 / $\pi_{0.5}$ VLA policies.

Chanyeok Choi, Youngmoon Lee

angledsugar.github.io/projects/0_cpd

Poisoning Attacks on Multi-Agent Reinforcement Learning Systems

Chanyeok Choi, Jaehwan Cho, Youngmoon Lee

HAPTics: Human Action Prediction in Real-time via Pose Kinematics

Niaz Ahmad, Saif Ullah, Jawad Khan, Chanyeok Choi, Youngmoon Lee

Causes and Fixes of Unexpected Drone Shutoffs

Hojun Choi, Chanyeok Choi, Youngmoon Lee

Snapbot: Enabling Dynamic Human Robot Interactions for Real-Time Computational Photography

Chanyeok Choi, Youngmoon Lee

SSK-DNN: Semantic and Sentiment Knowledge for Incremental Text Sentiment Classification

Jawad Khan, Niaz Ahmad, Chanyeok Choi, Saif Ullah, Gyu Rin Kim, Youngmoon Lee

Awards

Information & Communication Planning and Evaluation President Award (2nd place) — SW Talent Festival

Jan 2023

AnimeGAN Filter Photographer is a manipulator that frames and captures stylized portraits under safety-RL constraints.

Capstone Design Award (3rd place) — Hanyang University

Jan 2021

Guide Robot Dog is a full-stack quadruped robot motivated by guide dogs, intended to assist visually impaired pedestrians.

Teaching

Lead Teaching Assistant — Robot Vision System: Hanyang University, Fall 2022 & Fall 2023. Lead other graduate TAs for a class of 30–40 students; developed an autonomous robot training and vision environment in Unity. [\[Materials\]](#)

Teaching Assistant — Robot Learning: Hanyang University, Spring 2022 & Spring 2023. Class of 40–50 students. Designed a Pac-Man RL project and an ultrasonic-sensor self-driving car kit for real-world maze escape. [\[Materials\]](#)

Skills

Robot Learning

Perception

Engineering

Languages

Korean

Native speaker

English

Professional working proficiency

Interests

Research Interests